SEQUENCES OF OPERATION

PART I

PART II (Not Applicable)

PART III

3.01 GENERAL

A. The existing Building Automation System controls and equipment are as manufactured SIEMENS. New controls and equipment shall be provided and installed to monitor and control the indoor environment of the new facility expansion. Contractor shall take responsibility to provide for extension/expansion of the existing BAS system and full integration into the existing Facility Management System for Life Safety, Access Control, Energy Management, Utility Monitoring, Temperature Control, and Perimeter Intrusion Detection.

B. The sequences of operations presented are taken from the original construction documents and modified herein to include new equipment items. It is not the intent of these sequences to alter existing operations. Coordinate sequences with building facilities manager prior to commencement of work.

C. Sequences of Operations provided for existing equipment is provided for informational and coordination purposes.

3.02 VOLUME BOXES

A. Variable Volume Boxes (VAV):

1. A direct digital application specific controller shall modulate the box damper to maintain a variable air volume independent of the duct static pressure to satisfy the room temperature setpoint.
2. A room temperature sensor with an integral setpoint adjustment shall be provided for each VAV box. The temperature adjustment span shall be set through the Operator workstation.
3. In the event that the room temperature drops 2°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # __(room name)__ below setpoint" and the area served by this VAV on the graphical floor plan shall be shaded blue. In the event that the room temperature rises 2°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # __(room name)__ above setpoint" and the area served by this VAV on the graphical floor plan shall be shaded red. When the room temperature is within its predetermined (adjustable) temperature range the background shall remain white.
4. Initiate a historical log on the room temperature. Plot the temperatures at 30 minute intervals. Store the samples to individual computer files "Room#___" on the Operator Workstation hard drive.

B. Fan Powered VAV Boxes (FPB)

1. A direct digital application specific controller shall modulate the box damper and start/stop the internal fan to maintain a variable air volume independent of the duct static pressure to satisfy the room temperature setpoint.
2. A room temperature sensor with an integral setpoint adjustment shall be provided for each VAV box. The temperature adjustment span shall be set through the Operator workstation.
3. Upon a call for full cooling the primary air valve supplies maximum flow and the fan is off. At reduced cooling demand the primary air flow is reduced and the fan is energized at minimum fan flow. On a further reduction of cooling demand, the primary air flow decreases while the fan flow is proportionately increased. On a call for heat the primary air flow is at minimum while the fan flow is at its maximum setpoint. If additional heat is required, an electric reheat coil is energized in stages.
4. In the event that the room temperature drops 2°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # __(room name)__ below setpoint" and the area served by this VAV on the graphical floor plan shall be shaded blue. In the event that the room temperature rises 2°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # __(room name)__ above setpoint" and the area served by this VAV on the graphical floor plan shall be shaded red. When the room temperature is within its predetermined (adjustable) temperature range the background shall remain white.
5. Initiate a historical log on the room temperature. Plot the temperatures at 30 minute intervals. Store the samples to individual computer files "Temperature Room #___" on the Operator Workstation hard drive.

3.03 AIR HANDLING UNITS

A. Energy Recovery Ventilators:

1. The air handling unit shall be started and stopped by the facility management system through software interlocks with their respective air handling unit supply fans.
   ERV-1 – AHU 1-1, 1-2, 1-3, 2-1, 2-2, 3-1, 3-2
   ERV-2 – AHU 1-4, 2-3, 2-4, 3-3, 3-4
   ERV-3 – AHU 1-5, 1-6, 2-5, 2-6, 3-5, 3-6
   ERV-4 – AHU 1-8, 1-9, 2-7, 2-8, 3-7, 3-8
   ERV-5 – AHU 1-10, 1-11, 2-8, 2-10, 3-9, 3-10
2. A software signal through the ERV supply fan VFD shall provide positive run indication to the system operator. In the event that the supply fan is commanded to start, but fails to start a system advisory shall be provided to the system operator, "Energy Recovery Ventilator #___ supply fan off normal". In the event that the fan is commanded to stop, but the fan continues to run a system advisory shall be provided to the system operator, "Energy Recovery Ventilator #___ supply fan located in room #___ off normal".
3. The total energy wheel motor and exhaust fan shall be commanded to start whenever the supply fan is energized. A software signal through the exhaust fan VFD shall provide positive run indication to the system operator. In the event that the exhaust fan is commanded to start, but fails to start a system advisory shall be provided to the system operator, "Energy Recovery Ventilator #___ exhaust fan off normal". In the event that the fan is commanded to stop, but the fan continues to run a system advisory shall be provided to the system operator, "Energy Recovery Ventilator #___ exhaust fan off normal".
4. The outside air shall be modulated to maintain a predetermined (adjustable) outside air volume, the exhaust air shall be modulated to track the outside air volume less a predetermined (adjustable) pressurization offset. Where applicable, the combination fire/smoke dampers shall open when the Energy Recovery ventilator is commanded to start. The dampers shall close when the Energy Recovery Ventilator is stopped. The facility management system shall monitor the damper positions. In the event the dampers are commanded to open, but fail to open, a service advisory shall be provided to the system operator, "Dampers on ERV#___, commanded to open, but failed". In the event the dampers are commanded to close, but fail to close, a service advisory shall be provided to the system operator, "Dampers on ERV#___, commanded to close, but failed".
5. If both air handling units on a floor are running, energize the toilet exhaust and the general exhaust. If one air handling unit on a floor is running energize the toilet exhaust and close the general exhaust. If both air handling units on a floor are off, de-energize the toilet exhaust and general exhaust. For ERV-1, the general exhaust on the first floor shall have three setpoints:
   - Closed when AHU 1-1 and AHU 1-3 are off. Maintain 2,660 CFM
   - when AHU 1-1 is running and AHU 1-2 is off, Maintain 6,180 CFM
   - when both AHU 1-1 and 1-2 are running.
6. A differential pressure sensor located on the bottom of the supply air riser (1st floor) shall modulate the VFD to maintain a predetermined (adjustable) static pressure setpoint. In the event that the supply static pressure drops .25" below the pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Supply static pressure on Energy Recovery Ventilator ___ below setpoint" and the supply static pressure sensor on the Energy Recovery Ventilator graphic shall change from green to yellow. In the event that the supply static pressure rises .25" above the static pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Supply static pressure on Energy Recovery Ventilator ___ above setpoint" and the supply static pressure sensor on the air handling unit graphic shall change from green to red.
7. A differential pressure sensor located at the top of the exhaust air riser (3rd floor) shall modulate the VFD to maintain a predetermined (adjustable) negative static pressure setpoint. In the event that the exhaust static pressure drops .25" below the pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Exhaust static pressure on Energy Recovery Ventilator ___ below setpoint" and the exhaust static pressure sensor on the Energy Recovery Ventilator graphic shall change from green to red. In the event that the exhaust static pressure rises .25" above the static pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Exhaust static pressure on Energy Recovery Ventilator ___ above setpoint" and the exhaust static pressure sensor on the Energy Recovery Ventilator graphic shall change from . green to yellow.
8. A differential pressure switch piped across the pre-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Pre-filter on Energy Recovery Ventilator #___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
9. A differential pressure switch piped across the final-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Final-filter on Energy Recovery Ventilator #___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
10. A differential pressure switch piped across the exhaust-filter bank shall provide a service advisory to the system in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Exhaust-filter on Energy operator recovery Ventilator #___ dirty" and the graphical representation of the filter shall change color from green to red.
11. The Application Specific Controller shall modulate the normally closed 2-way chilled water valve to maintain a predetermined (adjustable) discharge air temperature setpoint, as sensed by the temperature sensor mounted in the supply air duct. In the event that the supply temperature drops 5°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Supply air temperature on Energy Recovery Ventilator #__ below setpoint" and the supply air temperature sensor on the air handling unit graphic shall change from green to blue. In the event that the supply temperature rises 5°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Supply air temperature on Energy Recovery Ventilatorr# ___ above setpoint and the supply air temperature sensor on the Energy Recovery Ventilator graphic shall change from green to red.
12. The unit shall be disabled by a signal from the fire alarm system. If the unit is stopped through the fire alarm system, a service advisory shall be provided, "Energy Recovery Ventilatorr# ___ stopped from Fire Alarm System". Provide specific advisory as to which automatic fire alarm device has stopped the ERV.
13. Normal operation shall resume when the F/A has been reset.
14. A low limit controller(s) with a 20' capillary placed in front of the cooling coil shall open the chilled water valve if freezing temperature is present. A service advisory shall be provided at the Operator Workstation, "Low Limit Alarm on Energy Recovery Ventilator #___ reset manually to restart unit".
15. In the event that the incoming power to the VFD(s) is outside preset (adjustable) operating parameters the Energy Recovery Ventilator VFD shall stop and a service advisory shall be provided to the system operator, "Incoming Power to the VFD on ERV # __ located in room ___ off normal".
16. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page (beeper message) with the corresponding alarm message:
   - Fire Alarm, Alarm Condition
   - Fire Alarm, Trouble Condition
   - Low Entering Air Temperature condition
   - Low Supply Air Temperature Condition
   - High Supply Air Temperature Condition
   - High Supply Air Static Pressure Condition
   - Low Supply Air Static Pressure Condition
   - Low Exhaust Air Static Pressure Condition
   - High Exhaust Air Static Pressure Condition
   - Air Handling Unit Supply Fan Off Normal
   - Dirty Filter Condition
   - Damper position off normal
   - Supply Fan VFD Power - Off Normal
   - Exhaust Fan VFD Power - Off Normal
   - Provide all other indication points as described in the points list.
17. Initiate a trend log on the supply air temperature, supply air static pressure, exhaust air static pressure, and the entering outside air temperature. Plot the samples at 30 minute intervals. Store the samples to individual ERV files "(sample name) AHU## ___" on the Operator Workstation hard drive.
18. Initiate a run status log for each fan. Store the sample's time and date of each start and stop. Store the samples to a file "Run ERV#__" on the Operator Workstation hard drive.

B. Variable Air Volume Air Handling Units:

1. The air handling unit shall be started and stopped by the facility management system through software schedules. Occupants shall be provided with the ability to override the automatic schedule during after hours through their desk top telephone. The system shall allow a two hour (adjustable) override by event.
   Provide a software interlock as follows: When AHU 1-3 is started, AHU 1-2 shall be started. Whenever AHU 1-2 is started ERV-1 shall be started and toilet exhaust shall be provided as scheduled.
2. A software signal through the supply fan VFD shall provide positive run indication to the system operator. In the event that the supply fan is commanded to start, but fails to start a system advisory shall be provided to the system operator, "Air Handling Unit #___ supply fan located in room #___ off normal". In the event that the fan is commanded to stop, but the fan continues to run a system advisory shall be provided to the system operator, "Air Handling Unit #___ supply fan located in room #___ off normal".
3. The outside air volume control device shall be energized whenever the AHU supply fan is started. The volume control device shall measure the entering air volume and shall maintain a preset (adjustable) air volume setpoint. When the AHU is off the control device shall be closed. In the event that the air volume drops 200 CFM below its setpoint a service advisory shall be provided to the system operator "Entering outside air volume on AHU #___ Low". In the event that the air volume rises 200 CFM above its setpoint a service advisory shall be provided to the system operator "Entering outside air volume on AHU #___ High".
4. A mixed air temperature sensor shall modulate the outside air preheat coil to maintain a predetermined (55°F adjustable)temperature setpoint. In the event that the mixed air temperature drops 5°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Mixed air temperature on AHU# located in room ___below setpoint" and the mixed air temperature sensor on the air handling unit graphic shall change from green to blue. In the event that the mixed air temperature rises 10°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Mixed air temperature on AHU located in room ___above setpoint" and the supply air temperature sensor on the air handling unit graphic shall change from green to red.
5. A differential pressure sensor located near the end of the supply air shall modulate the VFD to maintain a predetermined (adjustable) static pressure setpoint. In the event that the supply static pressure drops 0.5" below the pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Supply static pressure on AHU located in room ___below setpoint" and the supply static pressure sensor on the air handling unit graphic shall change from green to yellow. In the event that the supply static pressure rises .5" above the static pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Supply static pressure on AHU # located in room ___above setpoint" and the supply static pressure sensor on the air handling unit graphic shall change from green to red.
6. A differential pressure switch piped across the pre-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Pre-filter on AHU #___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
7. A differential pressure switch piped across the Final-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Final-filter on AHU #___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
8. The Application Specific Controller shall modulate the normally closed 2-way chilled water valve to maintain a predetermined (adjustable) discharge air temperature reset schedule, as sensed by the temperature sensor mounted in the supply air duct and reset by the return air temperature sensor. In the event that the supply temperature drops 5°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Supply air temperature on AHU# ___ located in room ___below setpoint" and the supply air temperature sensor on the air handling unit graphic shall change from green to blue. In the event that the supply temperature rises 5°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Supply air temperature on AHU# located in room ___above setpoint" and the supply air temperature sensor on the air handling unit graphic shall change from green to red.
9. The unit shall be disabled by a signal from the fire alarm system. If the unit is stopped through the fire alarm system, a service advisory shall be provided, "AHU # ___ located in room #___ stopped from Fire Alarm System". Provide specific advisory as to which automatic fire alarm device has stopped the AHU.
10. Normal operation shall resume when the F/A has been reset.
11. In the event that the incoming power to the VFD is outside preset (adjustable) operating parameters the air handling unit VFD shall stop and a service advisory shall be provided to the system operator, "Incoming Power to the VFD on AHU # ___ located in room ___ off normal".
12. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page (beeper message) with the corresponding alarm message:
   - Fire Alarm, Alarm Condition
   - Fire Alarm, Trouble Condition
   - Low Entering Outside Air Flow condition
   - High Entering Outside Air Flow Condition
   - Low Supply Air Temperature Condition
   - High Supply Air Temperature Condition
   - Low Return Air Temperature Condition
   - High Return Air Temperature Condition
   - Low Mixed Air Temperature Condition
   - High Supply Air Static Pressure Condition
   - Low Supply Air Static Pressure Condition
   - Air Handling Unit Supply Fan Off Normal
   - Dirty Filter Condition
   - Entering VFD Power - Off Normal
   - Provide all other indication points as described in the points list.
13. Initiate a trend log on the supply air temperature, supply air static pressure, mixed air temperature, return air temperature, and the outside air flow rate. Plot the samples at 30 minute intervals. Store the samples to individual AHU files "(sample name) AHU # ___" on the Operator Workstation hard drive.
14. Initiate a run status log for each fan. Store the sample's time and date of each start and stop. Store the samples to a file "Run AHU#__" on the Operator Workstation hard drive.

C. Constant Volume Air Handling Units:

1. The air handling unit shall be started and stopped by the facility management system through software schedules. The units shall run continuously.
2. A current sensor shall provide positive run indication to the system operator. In the event that the supply fan is commanded to start, but fails to start a system advisory shall be provided to the system operator, "Air Handling Unit #___ supply fan located in room #___ off normal". In the event that the fan is commanded to stop, but the fan continues to run a system advisory shall be provided to the system operator, "Air Handling Unit #___ supply fan located in room #___ off normal".
3. The outside air damper shall open whenever the AHU supply fan is started. The damper shall close when the air handling unit is stopped.
4. A differential pressure switch piped across the pre-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Pre-filter on AHU # ___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
5. A differential pressure switch piped across the final-filter bank shall provide a service advisory to the system operator in the event that the differential pressure increases above a predetermined (adjustable) setpoint, "Final-filter on AHU #___ located in room #___ dirty" and the graphical representation of the filter shall change color from green to red.
6. The Application Specific Controller shall modulate the normally closed 2-way chilled water valve to maintain a predetermined (adjustable) return air temperature setpoint, as sensed by the temperature sensor mounted in the return air duct. In the event that the return temperature drops 5°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Return air temperature on AHU # ___ located in room ___below setpoint" and the return air temperature sensor on the air handling unit graphic shall change from green to blue. In the event that the return temperature rises 5°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Return air temperature on AHU ___ located in room ___above setpoint and the return air temperature sensor on the air handling unit graphic shall change from green to red.
7. The unit shall be disabled by a signal from the fire alarm system. If the unit is stopped through the fire alarm system, a service advisory shall be provided, "AHU # ___ located in room #___ stopped from Fire Alarm System". Provide specific advisory as to which automatic fire alarm device has stopped the AHU.
8. Normal operation shall resume when the F/A has been reset.
9. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page (beeper message) with the corresponding alarm message:
   - Fire Alarm, Alarm Condition
   - Fire Alarm, Trouble Condition
   - Low Return Air Temperature Condition
   - High Return Air Temperature Condition
   - Air Handling Unit Supply Fan Off Normal
   - Dirty Filter Condition
   - Provide all other indication points as described in the points list.
10. Initiate a trend log on the supply air temperature and return air temperature. Plot the samples at 30 minute intervals. Store the samples to individual AHU files "(sample name) AHU # ___" on the Operator Workstation hard drive.
11. Initiate a run status log for each fan. Store the sample's time and date of each start and stop. Store the samples to a file "Run AHU#__" on the Operator Workstation hard drive.

3.04 FAN COIL UNITS:

A. Fan Coil Units:

1. The fan coil units shall run continuously.
2. The Application Specific Controller shall control the normally closed 2-way chilled water valve to maintain a predetermined (adjustable) room temperature setpoint. The room temperature shall be monitored and controlled through the Operator Workstation. In the event that the room temperature drops 2T below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # ___ (room name)__ below setpoint and the area served by this FCU on the graphical floor plan shall be shaded blue. In the event that the room temperature rises TF above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Room temperature in room # ___ (room name)__ above setpoint and the area served by this FCU on the graphical floor plan shall be shaded red. When the room temperature is within its predetermined (adjustable) temperature range the background shall remain white.
3. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page ("beeper message") with the corresponding alarm message:
   - Low Exhaust Air Flow Condition
   - High Exhaust Air Flow Condition
   - Provide all other indication points as described in the points list.

3.05 EXHAUST AIR

A. Toilet / General Exhaust:

1. Exhaust air shall be measured and controlled through an exhaust air control device.
2. The exhaust air control device shall activate when its associated air handling unit is started. The exhaust air device shall maintain a predetermined (adjustable) exhaust air volume as scheduled.
3. The device shall close when its associated air handling unit is stopped.
4. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page ("beeper message") with the corresponding alarm message:
   - Low Room Temperature Condition
   - High Room Temperature Condition
   - Provide all other indication points as described in the points list.

3.06 CHILLED WATER PLANT

A. Secondary Chilled Water Loop:

1. The secondary chilled water pumps shall be started to maintain a predetermined adjustable secondary chilled water loop differential pressure setpoint through the VFD when any chilled unit is operational. Provide an RS-232/RS-485 communication interface with each speed drive. Provide run status indication to the system operator. In the event that the pump is commanded to start but fails, a service advisory shall be provided to the system operator, "Secondary Chilled Water Pump #___ commanded to start but failed. If the chilled water pump is commanded off, but the pump continues to run, a service advisory shall be provided to the system operator "Secondary Chilled Water Pump #___ commanded to stop, but continued to run".
2. A differential pressure sensor piped across the chilled water supply and return lines at the furthest coil shall provide system differential pressure sensing and shall be used to control the variable speed drive (VFD) for the secondary chilled water loop. In the event that the differential pressure drops 2psi below the differential pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Chilled Water Pressure, below setpoint" and the differential pressure sensor on the graphic shall change from green to yellow. In the event that the differential pressure rises 2psi above the pressure setpoint, a service advisory shall be provided at the Operator Workstation, "Chilled water pressure, above setpoint" and the differential pressure sensor on the graphic shall change from green to red.
3. Provide software calculations to provide BTU calculations for instantaneous and BTUH (Ton/Hours) load calculation.
4. When any of the following conditions are present, as sensed by the facility management system, the system operator shall receive a alpha page ("beeper message") with the corresponding alarm message:
   - High Secondary Chilled Water Supply Temperature
   - Low Secondary Chilled Water Supply Temperature
   - High Secondary Chilled Water Pressure Low Secondary Chilled Water Pressure
   - Pump off Normal (each pump)
5. Initiate a trend log on the entering and leaving chilled water temperatures, flow and BTUH. Plot the temperatures at 15 minute intervals. Store the samples to a file "SCHW" on the Operator Workstation hard drive.
6. Initiate a run status log for each pump. Store the sample's time and date of each start and stop. Store the samples to a file "SCHWP" on the Operator Workstation hard drive.

B. Primary Chilled Water Loop:

1. On a weekly basis, the FMS shall index the chillers into their lead and lag positions to maintain equal runtime setpoints of the machines.
2. When a chiller is indexed to the run mode by the FMS the chiller shall start its dedicated primary chilled water pump.
3. A current sensor wired across the load side of th

B. Primary Chilled Water Loop:

1. On a weekly basis, the FMS shall index the chillers into their lead and lag positions to maintain equal runtime setpoints of the machines.
2. When a chiller is indexed to the run mode by the FMS the chiller shall start its dedicated primary chilled water pump.
3. A current sensor wired across the load side of the pump starter shall provide pump run indication.
4. When flow has been proven the chiller shall be started. Operating and fault conditions shall be logged at the Operator Workstation through an RS-232/RS-485 interface with the chiller manufacturer's (Open Protocol) control panel (see the specification for points list).
5. The lag chiller shall be started when the secondary flow exceeds 50% (adjustable) of its rated total flow capacity. The lag chiller shall be stopped when the lead chiller amperage draw is below 40% (adjustable) its rated full load amperage.
6. Chiller operating conditions shall be trended and stored to the HVAC OWS computer hard drive. Operating trends shall be provided on an 15 minute interval for entering and leaving chilled water temperatures, entering and leaving condensor water temperatures, kilowatt draw and voltage input. Store the samples to a file "PCHW" on the Operator Workstation hard drive.

C. Condensor Water:

1. When a chiller is indexed to the run mode by the FMS the chiller shall start its dedicated condensor water pump.
2. A current sensor wired across the load side of the pump starter shall provide pump run indication.
3. When a pump is started, it's cooling tower cell isolation valve shall open and the fan shall be enabled to run.
4. A condensor water supply temperature sensor shall modulate the VFD on the cooling towers to maintain a predetermined (adjustable) condensor water supply temperature setpoint. An RS-232/RS-485 signal from the VFD shall provide drive status and run indication to the system operator. In the event that the condensor water supply temperature drops 4°F below the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Condensor Water Supply temperature, below setpoint" and the condensor water supply temperature sensor on the graphic shall change from green to blue. In the event that the condensor water supply temperature rises 4°F above the temperature setpoint, a service advisory shall be provided at the Operator Workstation, "Condensor Water Supply temperature, above setpoint" and the condensor water supply temperature sensor on the graphic shall change from green to red.

Building Inspection Division
City of Lakeland
THESE PLANS HAVE BEEN REVIEWED FOR COMPLIANCE WITH THE CURRENT EDITION OF THE FLORIDA BUILDING CODE
Date: 11/15/19
Permit Number: BLD19-06035

AECOM
BATSON-COOK CONSTRUCTION

PROJECT

PUBLIX HEADQUARTERS BUILDING ADDITION AND PARKING LOT EXPANSION

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JOHN T. RECTOR, PROFESSIONAL ENGINEER, STATE OF FLORIDA, LICENSE NO. 53292
08/08/19

John T Rector
John T Rector
2019.08.22
15:46:35 -04'00'

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY JOHN T. RECTOR, PROFESSIONAL ENGINEER, STATE OF FLORIDA, LICENSE NO. 53292 ON THE DATE ADJACENT TO THE SEAL. PRINTED COPIES OF THIS DOCUMENT MUST BE VERIFIED ON ANY ELECTRONIC COPIES.

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ISSUE/REVISION

I/R	DATE	DESCRIPTION

PROJECT NUMBER

60601397

SHEET TITLE

HVAC CONTROLS

SHEET NUMBER

M6-0-01

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